

Introduction to Category Theory

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Definition

A *category* $\mathcal{C}$ consists of the following data:

- A collection $\text{Ob}(\mathcal{C})$ of *objects*;
- For each pair of objects A, B , a collection $\text{hom}_{\mathcal{C}}(A, B)$ whose elements are called *morphisms* or *arrows*;
- Composition: Given two morphisms $f: A \rightarrow B$ e $g: B \rightarrow C$, there exists a morphism $g \circ f: A \rightarrow C$;

Such that:

- Composition is associative;
- For each object A , there exists a morphism $1_A: A \rightarrow A$ called the *identity morphism* that acts as an identity for composition;

Notation:

We write $A \in \mathcal{C}$ for $A \in \text{Ob}(\mathcal{C})$
 $\text{hom}_{\mathcal{C}}(A, B)$, $\text{hom}(A, B) \in \mathcal{C}(A, B)$

Examples

- **Set, Grp, Mon, Ring, Vect $\mathbb{R}$, Top**
- **N**
- Preorder / Partial order $(X, \leq)$
- **Mat($\mathbb{R}$)**
- Monoid M / group G

Types of Morphisms

Definition

Let $\mathcal{C}$ be a category. A morphism $f : A \rightarrow B$ is said to be an *isomorphism* or *invertible*, if there exists a morphism $g : B \rightarrow A$ such that $gf = 1_A$ e $fg = 1_B$.

If that is the case, we say that the objects A and B are *isomorphic*, which we denote by $A \cong B$.

Note: if f is an isomorphism, then its inverse is unique, so we use the notation f^{-1} .

Definition

A group G is a category with only one object where every morphism is an isomorphism.

Note: A category where every morphism is an isomorphism is called a *groupoid*.

Monomorphisms

Proposition

Let X, Y be sets and $f : X \rightarrow Y$ a function. The following are equivalent:

- f is injective;
- Given functions g, h :

$$Z \begin{array}{c} \xrightarrow{g} \\ \xrightarrow{h} \end{array} X \xrightarrow{f} Y$$

If $fg = fh$, then $g = h$;

- There exists a function $w : Y \rightarrow X$ such that $wf = 1_X$.

Definition

Let $\mathcal{C}$ be a category. A morphism $f : A \rightarrow B$ is a **monomorphism** or **mono** if, for any morphisms $g, h : C \rightrightarrows A$, the equality $fg = fh$ implies that $g = h$.

Epimorphisms

Proposition

Let X, Y be sets and $f : X \rightarrow Y$ a function. The following are equivalent:

- f is surjective;
- Given two functions g, h :

$$X \xrightarrow{f} Y \begin{matrix} \xrightarrow{g} \\ \xrightarrow{h} \end{matrix} Z$$

If $gf = hf$, then $g = h$;

- There exists a function $w : Y \rightarrow X$ such that $fw = 1_Y$.

Definition

Let $\mathcal{C}$ be a category. A morphism $f : A \rightarrow B$ is an *epimorphism* or *epic* if, for any morphisms $g, h : B \rightrightarrows C$, $gf = hf$ implies $g = h$.

Proposition

Let $\mathcal{C}$ be a category. If f is an isomorphism, then f is both a monomorphism and an epimorphism.

Note: f being a monomorphism + epimorphism does not imply that f is an isomorphism.

A category where this happens is called *balanced*.

Terminal and initial objects

Definition

Let $\mathcal{C}$ be a category. An object A is called:

- *terminal* if for every object B there exists exactly one morphism $B \rightarrow A$;
- *initial* if for every object B there exists exactly one morphism $A \rightarrow B$.

Proposition

Let $\mathcal{C}$ be a category. Then:

- *If $\mathcal{C}$ has a terminal object, then it is unique up to isomorphism;*
- *If $\mathcal{C}$ has an initial object, then it is unique up to isomorphism;*

Note: the terminal object is usually denoted by 1 and the initial object by 0 or $\emptyset$.

Opposite Category

Definition

Let $\mathcal{C}$ be a category. The *opposite category* of $\mathcal{C}$, denoted by $\mathcal{C}^{\text{op}}$, is given by:

- The objects of $\mathcal{C}^{\text{op}}$ are the objects of $\mathcal{C}$;
- The morphisms $\mathcal{C}^{\text{op}}$ are the same as the morphisms in $\mathcal{C}$, but we 'flip' the direction. Formally:

$$\text{hom}_{\mathcal{C}^{\text{op}}}(A, B) := \text{hom}_{\mathcal{C}}(B, A)$$

- Composition is given by $g \circ_{\text{op}} f = f \circ g$;

Proposition

Let $\mathcal{C}$ be a category. Then:

- An object A is terminal [initial] in $\mathcal{C}$ iff A is initial [terminal] in $\mathcal{C}^{\text{op}}$;
- A morphism f is mono [epic] in $\mathcal{C}$ iff f is epic [mono] in $\mathcal{C}^{\text{op}}$.

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. We define $\mathcal{C} \times \mathcal{D}$ as being the category given by:

- The objects are pairs (A, B) , where $A \in \mathcal{C}$ e $B \in \mathcal{D}$. In other words $\text{Ob}(\mathcal{C} \times \mathcal{D}) = \text{Ob}(\mathcal{C}) \times \text{Ob}(\mathcal{D})$;
- A morphism $(A, B) \rightarrow (A', B')$ is given by a pair (f, g) where $f : A \rightarrow A'$ is a morphism in $\mathcal{C}$ and $g : B \rightarrow B'$ is a morphism in $\mathcal{D}$;

Composition of two morphisms:

$$(A, B) \xrightarrow{(f, g)} (A', B') \xrightarrow{(f', g')} (A'', B'')$$

is defined as $(f', g') \circ (f, g) = (f' \circ f, g' \circ g)$.

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ is given by:

- For each object $A \in \mathcal{C}$, an object $F(A)$ in $\mathcal{D}$;
- Given a morphism $f : A \rightarrow B$ in $\mathcal{C}$, a morphism $F(f)$ in $\mathcal{D}$:

$$\begin{array}{ccc} A & & F(A) \\ f \downarrow & \xrightarrow{F} & \downarrow F(f) \\ B & & F(B) \end{array}$$

Such that $F(1_A) = 1_{F(A)}$ for every object $A \in \mathcal{C}$, and

$F(f \circ g) = F(f) \circ F(g)$ for all composable morphisms $A \xrightarrow{g} B \xrightarrow{f} C$ in $\mathcal{C}$.

Functors II

Definition

Let $\mathcal{C} \xrightarrow{F} \mathcal{D} \xrightarrow{G} \mathcal{K}$ be functors. We define the composition $G \circ F : \mathcal{C} \rightarrow \mathcal{K}$ to be the functor given by:

- For any $A \in \mathcal{C}$, then $(G \circ F)(A) = G(F(A))$;
- For any morphism f in $\mathcal{C}$, then $(G \circ F)(f) = G(F(f))$.

Definition

We define **Cat** as being the category whose objects are categories, and morphisms are functors, where composition is defined as above.

Definition

Two categories $\mathcal{C}$ and $\mathcal{D}$ are said to be isomorphic if they are isomorphic as objects in **Cat**. In other words, if there are functors $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ such that $GF = 1_{\mathcal{C}}$ e $FG = 1_{\mathcal{D}}$.

Full and Faithful

Note: Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. For each pair of objects $A, B \in \mathcal{C}$, the functor F induces a map

$$\begin{aligned} F : \operatorname{hom}_{\mathcal{C}}(A, B) &\rightarrow \operatorname{hom}_{\mathcal{D}}(F(A), F(B)) \\ f &\mapsto F(f) \end{aligned} \tag{1}$$

Definition

A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is said to be:

- *Full*, if the map (1) is surjective;
- *Faithful*, if the map (1) is injective.

Proposition

Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. Then F is an isomorphism iff it is bijective on objects, and fully faithful.

Proposition

Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. Then:

- 1. F preserves isomorphisms;*
- 2. If F is fully faithful, then F reflects isomorphisms;*
- 3. If F is faithful, then F reflects monomorphisms and epimorphisms.*

Corollary

Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor and A, B objects in $\mathcal{C}$. If $A \cong B$, then $F(A) \cong F(B)$, where the converse is true if, for example, F is fully faithful.

Contravariant Functors

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *contravariant functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ is given by:

- For each object $A \in \mathcal{C}$, an object $F(A)$ in $\mathcal{D}$;
- Given a morphism $f : A \rightarrow B$ in $\mathcal{C}$, a morphism $F(f)$ in $\mathcal{D}$:

$$\begin{array}{ccc} A & & F(A) \\ f \downarrow & \xrightarrow{F} & \uparrow F(f) \\ B & & F(B) \end{array}$$

Such that $F(1_A) = 1_{F(A)}$ for every object $A \in \mathcal{C}$, and

$F(f \circ g) = F(g) \circ F(f)$ for all composable morphisms $A \xrightarrow{g} B \xrightarrow{f} C$ in $\mathcal{C}$.

Contravariant Functors II

Proposition

A *contravariant functor* from $\mathcal{C}$ to $\mathcal{D}$ is the same thing as a functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$.

Proposition

Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. Then F is also a functor from $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$.

Note: As such, a functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ is the same thing as a functor $F : \mathcal{C} \rightarrow \mathcal{D}^{\text{op}}$.

Proposition

The assignment $\mathcal{C} \mapsto \mathcal{C}^{\text{op}}$ and $F \mapsto F$ defines a functor **Cat** $\rightarrow$ **Cat**.

Natural Transformations

Definition

Let $F, G : \mathcal{C} \Rightarrow \mathcal{D}$ be two functors. A *natural transformation* $\alpha : F \Rightarrow G$ is given by, for each $C \in \mathcal{D}$, a morphism

$$\alpha_C : F(C) \rightarrow G(C)$$

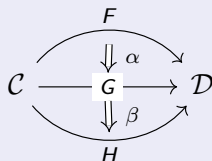
in $\mathcal{D}$, called the *components* of α in C , such that for each morphism $f : A \rightarrow B$ in $\mathcal{C}$, the following diagram called the *naturality square* of α commutes:

$$\begin{array}{ccc} F(A) & \xrightarrow{\alpha_A} & G(A) \\ F(f) \downarrow & & \downarrow G(f) \\ F(B) & \xrightarrow{\alpha_B} & G(B) \end{array}$$

Composition of Natural Transformations

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be two categories, and consider the following functors and natural transformations:



We define the composite $\beta \circ \alpha : F \Rightarrow H$ as the natural transformation with components given by the composition of α_A followed by β_A :

$$(\beta \circ \alpha)_A : F(A) \xrightarrow{\alpha_A} G(A) \xrightarrow{\beta_A} H(A)$$

so $(\beta \circ \alpha)_A = \beta_A \circ \alpha_A$ for each $A \in \mathcal{C}$.

Functor Categories

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be two categories. We define $[\mathcal{C}, \mathcal{D}]$ as being the category whose objects are functors from $\mathcal{C}$ to $\mathcal{D}$ and morphisms are natural transformations.

Definition

Two functors $F, G : \mathcal{C} \Rightarrow \mathcal{D}$ are said to be *naturally isomorphic*, written as $F \cong G$, if they are isomorphic as objects in $[\mathcal{C}, \mathcal{D}]$, i.e. if there are natural transformations $\alpha : F \Rightarrow G$ and $\beta : G \Rightarrow F$ such that $\alpha \circ \beta = 1_G$ and $\beta \circ \alpha = 1_F$. If this is the case, we say that α and β are *natural isomorphisms*.

Proposition

Let $F, G : \mathcal{C} \Rightarrow \mathcal{D}$ be two functors and $\alpha : F \Rightarrow G$ a natural transformation. Then α is a natural isomorphism iff for each $A \in \mathcal{C}$, the component $\alpha_A : F(A) \rightarrow G(A)$ is an isomorphism in $\mathcal{D}$.

Equivalent Categories

Note: Given two functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, we say that

$$F(A) \cong G(A), \text{ naturally in } A$$

to mean that F and G are naturally isomorphic.

Definition

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. We say that $\mathcal{C}$ and $\mathcal{D}$ are *equivalent*, denoted by $\mathcal{C} \simeq \mathcal{D}$, if there are functors $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ such that $F \circ G \cong 1_{\mathcal{D}}$ and $G \circ F \cong 1_{\mathcal{C}}$. If this is the case, then we say that F is an *equivalence* of categories.

A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is said to be *essentially surjective on objects*, if for each $B \in \mathcal{D}$, there is some $A \in \mathcal{C}$ such that $F(A) \cong B$.

Proposition

Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. Then F is an equivalence iff F is fully faithful and essentially surjective on objects.

Definition

Let $\mathcal{C}$ be a category. A *subcategory* $\mathcal{D}$ is given by:

- A subset $\text{Ob}(\mathcal{D}) \subseteq \text{Ob}(\mathcal{C})$ of objects;
- For each pair $A, B \in \text{Ob}(\mathcal{D})$, a subset of morphisms $\text{hom}_{\mathcal{D}}(A, B) \subseteq \text{hom}_{\mathcal{C}}(A, B)$

Such that $\mathcal{D}$ is itself a category.

Definition

A subcategory $\mathcal{D} \subseteq \mathcal{C}$ is said to be a *full subcategory* if $\text{hom}_{\mathcal{D}}(A, B) = \text{hom}_{\mathcal{C}}(A, B)$ for all $A, B \in \mathcal{D}$, i.e. if the inclusion functor $\iota : \mathcal{D} \hookrightarrow \mathcal{C}$ is full. (note that the inclusion functor is always faithful).

SubCategories II

Definition

A category $\mathcal{C}$ is said to be *skeletal*, if for any two objects $A, B \in \mathcal{C}$, if $A \cong B$ then $A = B$.

A *skeleton* of a category $\mathcal{C}$ is a skeletal subcategory whose inclusion functor is an equivalence.

Theorem

Every category has a skeleton. Furthermore, such skeleton is unique up to isomorphism.

Furthermore, two categories are equivalent if and only if they have isomorphic skeletons.

Dualities

An equivalence $\mathcal{C}^{\text{op}} \simeq \mathcal{D}$ is called a *duality* between the categories $\mathcal{C}$ and $\mathcal{D}$.

Examples:

- Stone duality:

$$\{\text{Boolean Algebras}\}^{\text{op}} \simeq \{\text{totally disconnected compact } T_2 \text{ spaces}\}$$

- Gelfand-Naimark duality:

$$\{\text{Commutative unital } C^*\text{-algebras}\}^{\text{op}} \simeq \{\text{compact } T_2 \text{ spaces}\}$$

- In algebraic geometry, given a ACF k :

$$\{\text{Affine Varieties over } k\}^{\text{op}} \simeq \{\text{f.g. } k\text{-algebras with no nontrivial nilpotents}\}$$

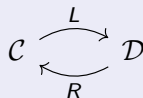
- Pontryagin Duality:

$$\{\text{Locally compact abelian top. Groups}\}^{\text{op}} \simeq \{\text{Locally compact abelian top. Groups}\}$$

Adjunctions

Definition

Consider categories and functors as follows:



We say that L is *left adjoint* to R and that R is *right adjoint* to L , written as $L \dashv R$, if there exists an isomorphism

$$\mathrm{hom}_{\mathcal{D}}(L(A), B) \cong \mathrm{hom}_{\mathcal{C}}(A, R(B)) \quad (2)$$

natural in $A \in \mathcal{C}$ and $B \in \mathcal{D}$.

An *adjunction* between L and R is a choice of isomorphism as in (2).

Adjunctions II

The bijection $\text{hom}_{\mathcal{D}}(L(A), B) \cong \text{hom}_{\mathcal{C}}(A, R(B))$ creates the assignment:

$$\begin{aligned} \left(L(A) \xrightarrow{g} B \right) &\mapsto \left(A \xrightarrow{\bar{g}} R(B) \right) \\ \left(L(A) \xrightarrow{\bar{f}} B \right) &\leftarrow \left(A \xrightarrow{f} R(B) \right) \end{aligned}$$

Where $\bar{\bar{f}} = f$. We call $\bar{f}$ the *transpose* of f . Naturality is the same thing as the following two conditions

$$\overline{\left(L(A) \xrightarrow{g} B \xrightarrow{q} B' \right)} = \left(A \xrightarrow{\bar{g}} R(B) \xrightarrow{R(q)} R(B') \right) \quad (3)$$

$$\overline{\left(A' \xrightarrow{p} A \xrightarrow{f} R(B) \right)} = \left(L(A') \xrightarrow{L(p)} L(A) \xrightarrow{\bar{f}} B \right) \quad (4)$$

That is, $\overline{q \circ g} = R(q) \circ \bar{g}$ and $\overline{f \circ p} = \bar{f} \circ L(p)$

Adjunctions III

Proposition

Consider adjunctions

$$\begin{array}{ccccc} \mathcal{C} & \xrightleftharpoons[L]{L} & \mathcal{D} & \xrightleftharpoons[R']{L'} & \mathcal{E} \end{array}$$

Then, $L' \circ L \dashv R \circ R'$. In fact, we have

$$\mathrm{hom}_{\mathcal{E}}(L'L(A), B) \cong \mathrm{hom}_{\mathcal{D}}(L(A), R'(B)) \cong \mathrm{hom}_{\mathcal{C}}(A, RR'(B))$$

Proposition

Left adjoints preserve initial objects and right adjoints preserve terminal objects.

Unit and Counit

Consider an adjunction

$$\mathcal{C} \begin{array}{c} \xrightarrow{L} \\ \perp \\ \xleftarrow{R} \end{array} \mathcal{D}$$

From the bijection $\text{hom}_{\mathcal{D}}(L(A), B) \cong \text{hom}_{\mathcal{C}}(A, R(B))$, we obtain the following bijection natural in $A \in \mathcal{C}$:

$$\text{hom}_{\mathcal{D}}(L(A), L(A)) \cong \text{hom}_{\mathcal{C}}(A, RL(A)) \quad (5)$$

In particular, for each $A \in \mathcal{C}$, we have a morphism

$$\left(A \xrightarrow{\eta_A} RL(A) \right) = \overline{\left(L(A) \xrightarrow{1} L(A) \right)}$$

Proposition

*Consider an adjunction as above. Then, the maps $\eta_A : A \rightarrow RL(A)$ form a natural transformation $\eta : 1_{\mathcal{C}} \Rightarrow RL$, known as the **unit** of the adjunction $L \dashv R$.*

Unit and Counit II

Consider an adjunction

$$\mathcal{C} \begin{array}{c} \xrightarrow{L} \\ \perp \\ \xleftarrow{R} \end{array} \mathcal{D}$$

From the bijection $\text{hom}_{\mathcal{D}}(L(A), B) \cong \text{hom}_{\mathcal{C}}(A, R(B))$, we obtain the following bijection natural in $B \in \mathcal{D}$:

$$\text{hom}_{\mathcal{D}}(LR(B), B) \cong \text{hom}_{\mathcal{C}}(R(B), R(B)) \quad (6)$$

In particular, for each $B \in \mathcal{D}$, we have a morphism

$$\left(LR(B) \xrightarrow{\varepsilon_B} B \right) = \overline{\left(R(B) \xrightarrow{1} R(B) \right)}$$

Proposition

Consider an adjunction as above. Then, the maps $\varepsilon_B : LR(B) \rightarrow B$ form a natural transformation $\varepsilon : LR \Rightarrow 1_{\mathcal{D}}$, known as the **counit** of the adjunction $L \dashv R$.

Whiskering of natural transformations

Consider the following:

$$\begin{array}{ccc} \mathcal{C} & \begin{array}{c} \xrightarrow{F} \\ \Downarrow \alpha \\ \xrightarrow{H} \end{array} & \mathcal{D} \xrightarrow{G} \mathcal{E} \end{array}$$

Then we have a composite:

$$\begin{array}{ccc} \mathcal{C} & \begin{array}{c} \xrightarrow{GF} \\ \Downarrow G\alpha \\ \xrightarrow{GH} \end{array} & \mathcal{E} \end{array}$$

With components

$$(G\alpha)_A := G(\alpha_A) : G(F(A)) \rightarrow G(H(A))$$

Whiskering of natural transformations II

Consider the following:

$$\mathcal{E} \xrightarrow{G} \mathcal{C} \begin{array}{c} \xrightarrow{F} \mathcal{D} \\ \Downarrow \alpha \\ \xrightarrow{H} \mathcal{D} \end{array}$$

Then we have a composite:

$$\mathcal{E} \begin{array}{c} \xrightarrow{FG} \mathcal{D} \\ \Downarrow \alpha G \\ \xrightarrow{HG} \mathcal{D} \end{array}$$

With components

$$(\alpha G)_A := \alpha_{G(A)} : F(G(A)) \rightarrow H(G(A))$$

Proposition

Consider an adjunction

$$\begin{array}{ccc} & L & \\ \mathcal{C} & \xrightarrow{\quad} & \mathcal{D} \\ & R & \end{array}$$

$\perp$

with unit η and counit ε . Then the following triangles commute (in $[\mathcal{C}, \mathcal{D}]$ and $[\mathcal{D}, \mathcal{C}]$ resp.)

$$\begin{array}{ccc} L & \xrightarrow{L\eta} & LRL \\ & \searrow 1_L & \downarrow \varepsilon L \\ & & L \end{array} \qquad \begin{array}{ccc} R & \xrightarrow{\eta R} & RLR \\ & \searrow 1_R & \downarrow R\varepsilon \\ & & R \end{array}$$

These two triangles are known as the *triangle identities*.

Proposition

Consider an adjunction

$$\begin{array}{ccc} \mathcal{C} & \begin{array}{c} \xrightarrow{L} \\ \perp \\ \xleftarrow{R} \end{array} & \mathcal{D} \end{array}$$

with unit η and counit ε , and let $A \in \mathcal{C}$ and $B \in \mathcal{D}$. Then, for any $g : L(A) \rightarrow B$,

$$\bar{g} = R(g) \circ \eta_A$$

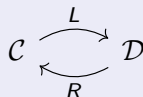
and for any $g : A \rightarrow R(B)$, we have that

$$\bar{f} = \varepsilon_B \circ L(f)$$

Adjunctions = Unit and Counit

Theorem

Consider two categories and functors



Then, there is a one-to-one correspondence between:

- 1. Adjunctions between L and R (with L being left adjoint);*
- 2. Pairs $(\eta : 1_{\mathcal{C}} \Rightarrow RL, \varepsilon : LR \Rightarrow 1_{\mathcal{D}})$ of natural transformations satisfying the triangle identities.*

Corollary

Using the same notation as the last theorem, there exists an adjunction $L \dashv R$ if and only if there are natural transformations $\eta : 1_{\mathcal{C}} \Rightarrow RL$ and $\varepsilon : LR \Rightarrow 1_{\mathcal{D}}$ satisfying the triangle identities.

Adjunctions IV

So we can think of adjunctions as:

1. Triples (L, R, Φ) with

$$\mathrm{hom}_{\mathcal{D}}(L(-), -) \stackrel{\Phi}{\cong} \mathrm{hom}_{\mathcal{C}}(-, R(-))$$

2. Quadruples $(L, R, \eta, \varepsilon)$ where $\eta : 1_{\mathcal{C}} \Rightarrow RL$ and $\varepsilon : LR \Rightarrow 1_{\mathcal{D}}$ satisfying the triangle identities.

Definition

A category $\mathcal{C}$ is said to be:

- *Small*, if the collection of all arrows is a set;
- *Locally small* if for every $A, B \in \mathcal{C}$, the collection $\text{hom}_{\mathcal{C}}(A, B)$ is a set;
- *Essentially small* if it is equivalent to a small category;

Definition

Let $\mathcal{C}$ be a locally small category, and fix an object A . We define a functor

$$h^A : \mathcal{C} \rightarrow \mathbf{Set}$$

as follows:

- For $B \in \mathcal{C}$, we define $h^A(B) = \text{hom}(A, B)$;
- For $f : B \rightarrow C$, then $h^A(f)$ is given by

$$\begin{aligned} h^A(f) : h^A(B) &\rightarrow h^A(C) \\ h &\mapsto f \circ h \end{aligned}$$

The notation $\text{hom}(A, -)$ is also common for h^A . We also sometimes denote $h^A(f)$ by $f \circ -$ or f_* .

Representables II

Definition

Let $\mathcal{C}$ be a locally small category and $F : \mathcal{C} \rightarrow \mathbf{Set}$ a functor. We say that F is *representable* if there exists some object $A \in \mathcal{C}$ such that $F \cong h^A$. A *representation* of F is a choice of object A and isomorphism between h^A and F .

Representables functors are sometimes just called *representables*.

Theorem

Any functor $R : \mathcal{C} \rightarrow \mathbf{Set}$ with a left adjoint is representable.

Representables III

Let $\mathcal{C}$ be a locally small category and consider a morphism $f : A \rightarrow B$. Then this induces a natural transformation:

The diagram illustrates a natural transformation h^f between two functors h^B and h^A from a category $\mathcal{C}$ to the category **Set**. The category $\mathcal{C}$ is on the left, and **Set** is on the right. Two curved arrows originate from $\mathcal{C}$ and point to **Set**: the top arrow is labeled h^B and the bottom arrow is labeled h^A . A vertical arrow labeled h^f points from the top arrow to the bottom arrow, representing the component of the natural transformation at each object of $\mathcal{C}$.

Where, for each $C \in \mathcal{C}$, the component of h^f is

$$(h^f)_C : h^B(C) \rightarrow h^A(C)$$
$$p \mapsto p \circ f$$

Definition

Let $\mathcal{C}$ be a locally small category. We define a functor

$$h^\bullet : \mathcal{C}^{\text{op}} \rightarrow [\mathcal{C}, \mathbf{Set}]$$

Where, given an object A , its image is h^A and, given an arrow f , its image is h^f .

Definition

Let $\mathcal{C}$ be a locally small category, and fix an object A . We define a functor

$$h_A : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$$

as follows:

- For $B \in \mathcal{C}$, we define $h_A(B) = \text{hom}(B, A)$;
- For $f : B \rightarrow C$, then $h_A(f)$ is given by

$$\begin{aligned} h_A(f) : h_A(C) &\rightarrow h_A(B) \\ h &\mapsto h \circ f \end{aligned}$$

The notation $\text{hom}(-, A)$ is also common for h_A . We also sometimes denote $h_A(f)$ by $- \circ f$ or f^* .

Definition

Let $\mathcal{C}$ be a locally small category and $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ a functor. We say that F is *representable* if there exists some object $A \in \mathcal{C}$ such that $F \cong h_A$. A *representation* of F is a choice of object A and isomorphism between h_A and F .

Recall: Functors of the form $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ are usually called *presheaves* on $\mathcal{C}$.

Representables VII

Let $\mathcal{C}$ be a locally small category and consider a morphism $f : A \rightarrow B$. Then this induces a natural transformation:

$$\begin{array}{ccc} \mathcal{C}^{\text{op}} & \begin{array}{c} \xrightarrow{h_A} \\ \Downarrow h_f \\ \xrightarrow{h_B} \end{array} & \mathbf{Set} \end{array}$$

Where, for each $C \in \mathcal{C}$, the component of h_f is

$$\begin{aligned} (h_f)_C : h_A(C) &\rightarrow h_B(C) \\ p &\mapsto f \circ p \end{aligned}$$

Definition

Let $\mathcal{C}$ be a locally small category. The *Yoneda embedding* is the functor

$$h_{\bullet} : \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$$

Where, given an object A , its image is h_A and, given an arrow f , its image is h_f .

Yoneda Lemma

Theorem (Yoneda)

Let $\mathcal{C}$ be a locally small category. Then

$$\mathrm{hom}_{[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]}(h_A, F) \cong F(A)$$

naturally in $A \in \mathcal{C}$ and $F \in [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$.

Corollary

For any locally small category $\mathcal{C}$, the Yoneda embedding

$$h_{\bullet} : \mathcal{C} \rightarrow [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$$

is fully faithful.

Yoneda Lemma II

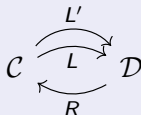
Corollary

Let $\mathcal{C}$ be a locally small category, and $A, B \in \mathcal{C}$. Then:

$$h_A \cong h_B \iff A \cong B \iff h^A \cong h^B$$

Corollary

Consider functors

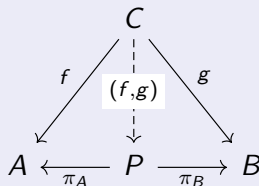


If $L \dashv R$ and $L' \dashv R$, then $L \cong L'$, i.e. the left adjoint of a fixed functor R are unique, up to isomorphism. The same goes for right adjoints.

Products

Definition

Let $\mathcal{C}$ be a category and consider objects A, B . A *product* of A and B (if it exists) is an object P and arrows $\pi_A : P \rightarrow A$ and $\pi_B : P \rightarrow B$ with the property that, given any arrows $f : C \rightarrow A$ and $g : C \rightarrow B$, then there exists a unique arrow $(f, g) : C \rightarrow P$ such that the following commutes:



Note: We will later see that the product of A and B , if it exists, is unique up to isomorphism, and so we use the notation $A \times B$.

Equalizers

Definition

Consider two arrows $f, g : A \rightarrow B$. An *equalizer* of f and g (if it exists) is an object E with a map $E \xrightarrow{i} A$ such that $fi = gi$ with the property that for any other $X \xrightarrow{w} A$ with $fw = gw$, there exists a unique map $X \xrightarrow{\bar{w}} E$ such that the following commutes

$$\begin{array}{ccccc} E & \xrightarrow{i} & A & \xrightarrow[f]{g} & B \\ \bar{w} \uparrow \text{---} & \nearrow w & & & \\ X & & & & \end{array}$$

Note: We will see later that the equalizer of a pair of parallel morphisms is unique, up to isomorphism.

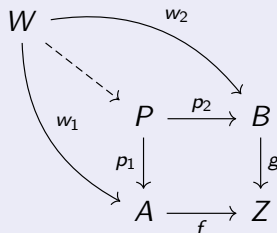
Lemma

*Every equalizer is a monomorphism. (A monomorphism which is the equalizer of some pair of parallel arrows is called a *regular monomorphism*)*

Pullbacks I

Definition

Let $\mathcal{C}$ be a category and consider two morphisms $f : A \rightarrow Z$ and $g : B \rightarrow Z$. A **pullback** of f and g (if it exists), is an object P with maps $P \xrightarrow{p_1} A$ and $P \xrightarrow{p_2} B$ such that $fp_1 = gp_2$, with the property that, for any other object W with maps $W \xrightarrow{w_1} A$ and $W \xrightarrow{w_2} B$ such that $fw_1 = gw_2$, there exists a unique map $W \rightarrow P$ such that the following commutes:



Note 1: We will see later that the pullback of two morphisms, if it exists, is unique up to isomorphism. As such, the pullback of $f : A \rightarrow Z$ and $g : B \rightarrow Z$ is sometimes denoted by $A \times_Z B$.

Note 2: The a commutative square

$$\begin{array}{ccc} P & \xrightarrow{p_2} & B \\ p_1 \downarrow & \lrcorner & \downarrow g \\ A & \xrightarrow{f} & Z \end{array}$$

is called a *pullback square* if P the pullback of f and g . The $\lrcorner$ is sometimes used to denote that the square is a pullback.

Proposition

Let

$$\begin{array}{ccc} P & \xrightarrow{p_2} & B \\ p_1 \downarrow & & \downarrow g \\ A & \xrightarrow{f} & Z \end{array}$$

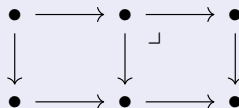
be a pullback square. Then:

- If g is a monomorphism, then p_1 is a monomorphism;
- If g is an isomorphism, then p_1 is an isomorphism;

The converse is not necessarily true, and the pullback of a epimorphism may not be epic.

Proposition

Consider the following diagram:



Assume that the inner right-hand side square is a pullback square, as signaled by the $\lrcorner$.

Then the outer square is a pullback square if and only if the inner left-hand side square is a pullback square.

Proposition

A map $X \xrightarrow{f} Y$ is a monomorphism if and only if the following is a pullback square:

$$\begin{array}{ccc} X & \xrightarrow{1} & X \\ 1 \downarrow & & \downarrow f \\ X & \xrightarrow{f} & Y \end{array}$$

Definition

Let $\mathcal{C}$ be a category and $\mathcal{I}$ a small category. A functor $D : \mathcal{I} \rightarrow \mathcal{C}$ is called a *diagram* in $\mathcal{C}$ of shape $\mathcal{I}$.

Definition

Consider a diagram $D : \mathcal{I} \rightarrow \mathcal{C}$. A *cone* of the diagram D is an object $A \in \mathcal{C}$ (called *vertex* of the cone) together with a family of morphisms

$$\left(A \xrightarrow{f_I} D(I) \right)_{I \in \mathcal{I}}$$

such that, for every maps $I \xrightarrow{u} J$ in $\mathcal{I}$, the following commutes:

A commutative triangle diagram illustrating the condition for a cone. At the top vertex is the object A . At the bottom-left vertex is the object $D(I)$. At the bottom-right vertex is the object $D(J)$. A morphism f_I points from A down to $D(I)$. A morphism f_J points from A down to $D(J)$. A morphism $D(u)$ points from $D(I)$ to $D(J)$ along the bottom edge.

Definition

Let $\mathcal{C}$ be a category and $D : \mathcal{I} \rightarrow \mathcal{C}$ a diagram of shape $\mathcal{I}$. A **limit** of D (if it exists), is a cone $\left(L \xrightarrow{p_I} D(I) \right)_{I \in \mathcal{I}}$ with the property that, for any other cone $\left(A \xrightarrow{f_I} D(I) \right)_{I \in \mathcal{I}}$, there exists a unique map $A \xrightarrow{\bar{f}} L$ such that the following commutes for each $I \in \mathcal{I}$

$$\begin{array}{ccc} A & \xrightarrow{\quad \bar{f} \quad} & L \\ & \searrow f_I \quad \swarrow p_I & \\ & D(I) & \end{array}$$

The maps p_I are called the **projections** of the limit and the cone $\left(L \xrightarrow{p_I} D(I) \right)_{I \in \mathcal{I}}$ is usually called a **limiting cone** for the diagram D .

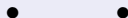
Proposition

Let $\mathcal{C}$ be a category and consider a diagram $D : \mathcal{I} \rightarrow \mathcal{C}$. Then, if this diagram has a limit in $\mathcal{C}$, that limit is unique up to isomorphism.

Note: As such, if the limit of the diagram $D : \mathcal{I} \rightarrow \mathcal{C}$ exists, we denote it by $\varprojlim_{\mathcal{I}} D$ or simply $\varprojlim D$ when $\mathcal{I}$ is implicit by context.

Definition (alternative def. of product)

Let $\mathcal{C}$ be a category and $A, B \in \mathcal{C}$. Let $\mathcal{I}$ be the discrete category with just two points



Let $D : \mathcal{I} \rightarrow \mathcal{C}$ be the diagram that maps one of the points in $\mathcal{I}$ to A and the other one to B . The limit, if it exists is called the *product* of A and B , denoted as $A \times B$.

Definition

Let $\mathcal{C}$ be a category and I be a set. Consider a set of objects $A_i \in \mathcal{C}$ indexed by I . Let $\mathcal{I}$ be the discrete category whose objects are the elements of I and consider the diagram $D : \mathcal{I} \rightarrow \mathcal{C}$ where $D(i) = A_i$. Then the limit $\varprojlim D$, if it exists, is called the **product** of the $\{A_i\}_{i \in I}$ and is denoted by $\prod_{i \in I} A_i$.

Notes

The product comes equipped with a projection map

$\pi_j : \prod_{i \in I} A_i \rightarrow A_j$ for each $j \in I$.

Given maps $f_i : C \rightarrow A_i$, then there exists a unique map $f : C \rightarrow \prod_{i \in I} A_i$ such that $\pi_i f = f_i$.

Definition (alternative def. of equalizer)

Let $\mathcal{C}$ be a category and consider two parallel maps $f, g : A \rightrightarrows B$. The *equalizer* of f and g is the limit of the following diagram:

$$(\bullet \rightrightarrows \bullet) \xrightarrow{D} \left(A \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} B \right)$$

Definition (alternative def. of terminal object)

Let $\mathcal{C}$ be a category and $D : \emptyset \rightarrow \mathcal{C}$ the empty diagram. Then a terminal object of $\mathcal{C}$, if it exists, is the limit of D .

Definition (alternative def. of pullback)

Let $\mathcal{C}$ be a category, and consider morphisms

$$\begin{array}{ccc} & B & \\ & \downarrow g & \\ A & \xrightarrow{f} & Z \end{array}$$

The *pullback* of f and g is the limit of the diagram:

$$\left(\begin{array}{ccc} & \bullet & \\ & \downarrow & \\ \bullet & \longrightarrow & \bullet \end{array} \right) \xrightarrow{D} \left(\begin{array}{ccc} & B & \\ & \downarrow g & \\ A & \xrightarrow{f} & Z \end{array} \right)$$

Proposition

Let $D : \mathcal{I} \rightarrow \mathbf{Set}$ be a diagram. Then, $\varprojlim D$ always exists and is given as a subset of $\prod_{I \in \mathcal{I}} D(I)$ by:

$$\varprojlim D = \{(x_I)_{I \in \mathcal{I}} : x_I \in D(I) \text{ and } Du(x_I) = x_J \text{ for all } I \xrightarrow{u} J \text{ in } \mathcal{I}\}$$

The limiting cone is given by the projections $\varprojlim D \xrightarrow{\pi_I} D(I)$, for each $I \in \mathcal{I}$.

Theorem

Let $\mathcal{C}$ be a category. Then:

- 1. If $\mathcal{C}$ has all products and equalizers, then it has all limits;*
- 2. If $\mathcal{C}$ has binary products, terminal object, and equalizers, then $\mathcal{C}$ has finite limits.*
- 3. If $\mathcal{C}$ has binary products, then $\mathcal{C}$ has equalizers if and only if $\mathcal{C}$ has pullbacks.*
- 4. If $\mathcal{C}$ has pullbacks and terminal object, then $\mathcal{C}$ has finite limits.*

Definition

A **cocone** on the diagram $D : \mathcal{I} \rightarrow \mathcal{C}$ is an object $A \in \mathcal{C}$ (the **vertex** of the cocone) together with a family of morphisms

$$\left(D(I) \xrightarrow{f_I} A \right)_{I \in \mathcal{I}}$$

such that, for all maps $I \xrightarrow{u} J$ in $\mathcal{I}$, the following commutes:

$$\begin{array}{ccc} D(I) & \xrightarrow{D(u)} & D(J) \\ & \searrow f_I & \swarrow f_J \\ & A & \end{array}$$

Definition

Let $\mathcal{C}$ be a category and $D : \mathcal{I} \rightarrow \mathcal{C}$ a diagram of shape $\mathcal{I}$. A **colimit** of D (if it exists), is a cocone $\left(D(I) \xrightarrow{p_I} L \right)_{I \in \mathcal{I}}$ with the property that, for any other cocone $\left(D(I) \xrightarrow{f_I} A \right)_{I \in \mathcal{I}}$, there exists a unique morphism $L \xrightarrow{\bar{f}} A$ such that the following commutes for each $I \in \mathcal{I}$

$$\begin{array}{ccc} & D(I) & \\ p_I \swarrow & & \searrow f_I \\ L & \xrightarrow{\quad \bar{f} \quad} & A \end{array}$$

The maps p_I are called the **coprojections** of the colimit and the cocone $\left(D(I) \xrightarrow{p_I} L \right)_{I \in \mathcal{I}}$ is usually called a **limiting cocone** for the diagram D .

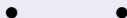
Proposition

Let $\mathcal{C}$ be a category and consider a diagram $D : \mathcal{I} \rightarrow \mathcal{C}$. Then, if this diagram has a colimit in $\mathcal{C}$, that colimit is unique up to isomorphism.

Note: As such, if the colimit of the diagram $D : \mathcal{I} \rightarrow \mathcal{C}$ exists, we denote it by $\varinjlim_{\mathcal{I}} D$ or simply $\varinjlim D$ when $\mathcal{I}$ is implicit by context.

Definition

Let $\mathcal{C}$ be a category and $A, B \in \mathcal{C}$. Let $\mathcal{I}$ be the discrete category with just two points



Let $D : \mathcal{I} \rightarrow \mathcal{C}$ be the diagram that maps one of the points in $\mathcal{I}$ to A and the other one to B . The colimit, if it exists is called the *coproduct* of A and B , denoted as $A + B$ or $A \coprod B$.

Note: Given a set of objects A_i for $i \in I$ in a category $\mathcal{C}$, we define their coproduct, denoted by $\sum_{i \in I} A_i$ or $\coprod_{i \in I} A_i$ is an analogous way as we did for products.

Coproducts II

In other words, the coproduct of A and B is an object $A + B$ with maps $i_A : A \rightarrow A + B$ and $i_B : B \rightarrow A + B$ forming a limiting cocone, i.e. with the property that for any $A \xrightarrow{f} C$ and $B \xrightarrow{g} C$, there exists a unique $A + B \xrightarrow{f+g} C$ such that the following commutes:

A commutative triangle diagram illustrating the universal property of the coproduct. The bottom row consists of three objects: A on the left, $A + B$ in the center, and B on the right. A solid arrow labeled i_A points from A to $A + B$, and a solid arrow labeled i_B points from B to $A + B$. Above $A + B$ is the object C . A solid arrow labeled f points from A to C , and a solid arrow labeled g points from B to C . A dashed arrow labeled $f+g$ points from $A + B$ to C , representing the unique map that makes the triangle commute.

Definition

Let $\mathcal{C}$ be a category and consider two parallel maps $f, g : A \rightrightarrows B$. The *coequalizer* of f and g is the colimit of the following diagram:

$$(\bullet \rightrightarrows \bullet) \xrightarrow{D} \left(A \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} B \right)$$

Coequalizers II

In other words, the coequalizer of f and g is an object E with a map $B \xrightarrow{p} E$ such that $pf = pg$ with the property that for any other $B \xrightarrow{w} X$ with $wf = wg$, there exists a unique map $E \xrightarrow{\bar{w}} X$ such that the following commutes

$$\begin{array}{ccccc} A & \xrightleftharpoons[f]{g} & B & \xrightarrow{p} & E \\ & & \searrow w & & \downarrow \bar{w} \\ & & & & X \end{array}$$

Lemma

*Every coequalizer is an epimorphism. (An epimorphism which is the coequalizer of some pair of parallel arrows is called a **regular epimorphism**).*

Pushouts

Definition

Let $\mathcal{A}$ be a category, and consider morphisms

$$\begin{array}{ccc} Z & \xrightarrow{g} & B \\ f \downarrow & & \\ A & & \end{array}$$

The *pushout* of f and g is the colimit of the diagram:

$$\left(\begin{array}{ccc} \bullet & \longrightarrow & \bullet \\ \downarrow & & \\ \bullet & & \end{array} \right) \xrightarrow{D} \left(\begin{array}{ccc} Z & \xrightarrow{g} & B \\ f \downarrow & & \\ A & & \end{array} \right)$$

Pushouts II

In other words, the pushout is an object P with maps $A \xrightarrow{p_1} P$ and $B \xrightarrow{p_2} P$ such that the following commutes:

$$\begin{array}{ccc} P & \xleftarrow{p_2} & B \\ \uparrow p_1 & & \uparrow g \\ A & \xleftarrow{f} & Z \end{array}$$

With the property that, for any other W with maps $A \xrightarrow{w_1} W$ and $B \xrightarrow{w_2} W$ with $w_1 f = w_2 g$, there exists a unique map $P \rightarrow W$ such that the following commutes:

$$\begin{array}{ccccc} & & W & \xleftarrow{w_2} & B \\ & \nwarrow \text{dashed} & \uparrow & & \uparrow g \\ & & P & \xleftarrow{p_2} & B \\ \uparrow w_1 & & \uparrow p_1 & & \\ A & \xleftarrow{f} & Z & & \end{array}$$

Pushouts III

If the following is a pushout square:

$$\begin{array}{ccc} P & \xleftarrow{p_2} & B \\ p_1 \uparrow & & \uparrow g \\ A & \xleftarrow{f} & Z \end{array}$$

We often use the notation $P = A \times_Z B$ or $P = A \coprod_Z B$

Proposition

Let

$$\begin{array}{ccc} P & \xleftarrow{p_2} & B \\ p_1 \uparrow & & \uparrow g \\ A & \xleftarrow{f} & Z \end{array}$$

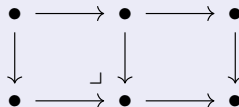
be a pushout square. Then:

- If g is an epimorphism, then p_1 is a epimorphism;
- If g is an isomorphism, then p_1 is an isomorphism;

The converse is not necessarily true.

Proposition

Consider the following diagram:



Assume that the inner left-hand side square is a pushout square, as signaled by the $\sqsubset$.

Then the outer square is a pushout square if and only if the inner right-hand side square is a pushout square.

Proposition

A map $Y \xrightarrow{f} X$ is an epimorphism if and only if the following is a pushout square:

$$\begin{array}{ccc} X & \xleftarrow{1} & X \\ 1 \uparrow & & \uparrow f \\ X & \xleftarrow{f} & Y \end{array}$$

Theorem

Let $\mathcal{C}$ be a category. Then:

- 1. If $\mathcal{C}$ has all coproducts and coequalizers, then it has all colimits;*
- 2. If $\mathcal{C}$ has binary coproducts, initial object, and coequalizers, then $\mathcal{C}$ has finite colimits.*
- 3. If $\mathcal{C}$ has binary coproducts, then $\mathcal{C}$ has coequalizers if and only if $\mathcal{C}$ has pushouts.*
- 4. If $\mathcal{C}$ has pushouts and initial object, then $\mathcal{C}$ has finite colimits.*

Proposition

Let $D : \mathcal{I} \rightarrow \mathbf{Set}$ be a diagram with $\mathcal{I}$ small. Then, $\varinjlim_{\mathcal{I}} D$ always exists and is given as a quotient:

$$\varinjlim_{\mathcal{I}} D = \left(\sum_{I \in \mathcal{I}} D(I) \right) / \sim$$

Where $\sim$ is the smallest equivalence relation on $\sum_{I \in \mathcal{I}} D(I)$ that identifies $x \sim (Du)(x)$ for all $I \xrightarrow{u} J$ in $\mathcal{I}$ and $x \in D(I)$.

The limiting cocone is given by projections into the quotient $D(I) \xrightarrow{\pi_I} \varinjlim_{\mathcal{I}} D$, for each $I \in \mathcal{I}$.